

Secant form / Quasi Newton Method [Optimized Method of]
- Simplifies the computation of Newton's Method.

Newton's Method, $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$

Here calculating two different functions.
Working With different functions & derivations,
it requires much computation in big data.

Goal: Simplify it and keep only in $f(x)$ function.

To do so, We need, [Showing Derivation]

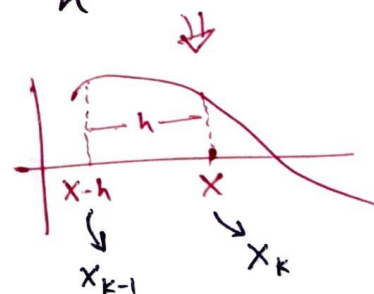
Backward difference, $f'(x) = \frac{f(x) - f(x-h)}{h}$

From the graph

Consider: $x_k = x$

$x_{k-1} = x - h$

So, $h = x - (x-h)$



Here, $f'(x) = \frac{f(x) - f(x-h)}{h} = \frac{f(x) - f(x-h)}{x - (x-h)}$
 $= \text{[plugging value of]} = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}$
 x_k, x_{k-1}

plugging $f'(x)$ value in Newton's Method:

$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} = x_k - \frac{f(x_k)}{\frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}}$

formula $\Rightarrow$

$\therefore x_{k+1} = x_k - \frac{(x_k - x_{k-1}) f(x_k)}{f(x_k) - f(x_{k-1})}$

Quasi Newton's Method

or, Secant Method formula;
$$X_{k+1} = X_k - \frac{(X_k - X_{k-1}) f(X_k)}{f(X_k) - f(X_{k-1})}$$

Ex. 1: $f(x) = x^2 - 5$; starting points, $X_0 = -0.5$,
 $X_1 = 0$
find root.

Ans.

Step 1 $X_0 = -0.5$, $X_1 = 0$

$$\underline{X_2} = X_1 - \frac{(X_1 - X_0) f(X_1)}{f(X_1) - f(X_0)} = 0 - \frac{(0 - (-0.5)) (0^2 - 5)}{(0^2 - 5) - \{(-0.5)^2 - 5\}}$$

Step 2

$$\underline{X_3} = X_2 - \frac{(X_2 - X_1) f(X_2)}{f(X_2) - f(X_1)} = -10 - \frac{(-10 - 0) \{(-10)^2 - 5\}}{\{(-10)^2 - 5\} - \{0^2 - 5\}}$$
$$= \underline{-0.5}$$

If you continue it
you will get:

K^{th}	X_K
0	-0.5
1	0
2	-10
3	-0.5
4	-0.95
5	-3.13
6	-1.883
7	-2.269
8	-2.236
9	-2.236

At 8th iteration,
the X_K value remains
unchanged. Hence,

$X_8 = -2.236$
is Root!

L14 - Newton's Method for Nonlinear Systems

1. Newton's Method Formula

For a system of nonlinear equations:

$$F(x, y) = \begin{pmatrix} f_1(x, y) \\ f_2(x, y) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Newton's method is:

$$X_{k+1} = X_k - J(X_k)^{-1}F(X_k)$$

where

$$X_k = \begin{pmatrix} x_k \\ y_k \end{pmatrix}$$

and $J(X_k)$ is the Jacobian matrix.

$$J(x, y) = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix}$$

2. Short Derivation

Using Taylor expansion around the current point X_k :

$$F(X_{k+1}) \approx F(X_k) + J(X_k)(X_{k+1} - X_k)$$

At the root:

$$F(X_{k+1}) = 0$$

Therefore,

$$0 = F(X_k) + J(X_k)(X_{k+1} - X_k)$$

Rearranging:

$$J(X_k)(X_{k+1} - X_k) = -F(X_k)$$

Multiplying by $J(X_k)^{-1}$:

$$X_{k+1} - X_k = -J(X_k)^{-1}F(X_k)$$

Hence,

$$\boxed{X_{k+1} = X_k - J(X_k)^{-1}F(X_k)}$$

3. Example

Solve:

$$f_1(x, y) = xy - y^3 - 1 = 0$$

$$f_2(x, y) = x^2y + y - 5 = 0$$

Initial guess:

$$(x_0, y_0) = (2, 3)$$

The function vector is:

$$F(x, y) = \begin{pmatrix} xy - y^3 - 1 \\ x^2y + y - 5 \end{pmatrix}$$

The Jacobian is:

$$J(x, y) = \begin{pmatrix} y & x - 3y^2 \\ 2xy & x^2 + 1 \end{pmatrix}$$

At $(2, 3)$:

$$F(2, 3) = \begin{pmatrix} -22 \\ 10 \end{pmatrix}$$

and

$$J(2, 3) = \begin{pmatrix} 3 & -25 \\ 12 & 5 \end{pmatrix}$$

The inverse Jacobian is:

$$J^{-1} = \frac{1}{315} \begin{pmatrix} 5 & 25 \\ -12 & 3 \end{pmatrix}$$

Using Newton's formula:

$$\begin{aligned} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} &= \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \frac{1}{315} \begin{pmatrix} 5 & 25 \\ -12 & 3 \end{pmatrix} \begin{pmatrix} -22 \\ 10 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 140/315 \\ 294/315 \end{pmatrix} \end{aligned}$$

Therefore:

$$\boxed{(x_1, y_1) = (1.55556, 2.06667)}$$

4. Practical Form

Instead of computing the inverse explicitly, solve:

$$J(X_k)\Delta X = -F(X_k)$$

then update:

$$\boxed{X_{k+1} = X_k + \Delta X}$$

This form is more stable and commonly used in numerical computation.